Paleomagnetism, structure, and longitudinal correlation of Middle Precambrian dykes from northwestern Ontario and Minnesota

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A primary magnetization about 2.1 Ga old is well preserved in the Kenora–Kabetogama dyke swarm in Minnesota and Ontario. Its direction, the mean for 12 dykes, incorporating results from 30 paleomagnetic sites, is 115.9°, −55.6°, yielding a paleomagnetic pole position of latitude 42.7°S, longitude 4.2°E (dm = 7.3°, dp = 5.2°).

The primary nature of the magnetization can be demonstrated using geochemistry to correlate individual dykes over distances that can exceed 300 km. Whereas the remanence direction varies *between* dykes, it remains constant *along* a dyke, sufficient proof that it was formed during initial cooling of the intrusion.

A weak, longitudinal increase in hydrous alteration, MgO content, and degree of magnetic overprinting is observed as the swarm is followed to the south and may reflect increasing proximity to a Precambrian continental margin that underwent rifting, sedimentation, and deformation during the time interval 2.2–1.6 Ga BP. Although the cause of the longitudinal changes in terms of margin evolution can only be surmised, the results illustrate the potential of dyke swarms as sensitive indicators of how shield terrains may respond to events occurring around their margins.

Une aimantation primaire âgée approximativement de 2,1 Ga est bien conservée dans le réseau de dykes Kenora–Kabetogama au Minnesota et en Ontario. Sa direction moyenne fournie par 12 dykes, incluant les résultats de 30 sites paléomagnétiques, est 115,9°, −55,6° et procure un pôle paléomagnétique de latitude 42,7°S, longitude 4,2°E (dm = 7,3°, dp = 5,2°).

La géochimie qui met en corrélation des dykes individuels sur des distances pouvant excéder 300 km vient corroborer la nature primaire de l'aimantation. Alors que la direction de rémanence varie *entre* les dykes, elle demeure toutefois constante *le long* du dyke, ce qui démontre de façon satisfaisante que son apparition date de la phase initiale du refroidissement de l'intrusion.

Un faible accroissement longitudinal de l'altération par l'eau, de la teneur en MgO et du degré de surimpression magnétique est observé en suivant le réseau de dykes vers le sud, ce qui peut refléter un rapprochement progressif vers la marge continentale précambrienne, où durant l'intervalle 2,2–1,6 Ga AvP se manifestaient rifting, sédimentation et déformation. Bien que pour le moment on soit limité à des hypothèses sur la cause de l'origine des changements longitudinaux en termes d'évolution de la marge, les résultats illustrent tout de même le potentiel d'utilisation des réseaux de dykes pour comprendre comment les terrains dans un bouclier répondent aux événements qui se produisent près de leurs marges.

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Introduction

In the southern Superior Province of the Canadian Shield, there is a profusion of dyke swarms emplaced approximately 2.0–2.2 Ga ago. These swarms trend at high angles to the southern margin of the Superior Province and gradually die out with increasing distance from this margin. They have been interpreted as "failed arm" remnants of major "hot spot" rifting activity that occurred along the southern margin from the Lake Superior region in the west to the Otish Mountains of Quebec in the east (Fahrig *et al.* 1984). Swarms of comparable age and origin also occur adjacent to the northern margin of the Superior Province but are less abundant.

Of the rift zones themselves little is known because they lie largely concealed beneath thick sedimentary wedges of Middle Precambrian strata that were subsequently deformed about 1.6–1.9 Ga ago in response to ocean closure (e.g., Baragar and Scoates 1981), forming major fold belts that now define the borders of the Superior craton.

The dyke swarms represent relatively fresh and undeformed products of these early rifting episodes and, through study of their radiometric age, paleomagnetism, and petrochemistry, may be used to reconstruct the breakup and dispersal of continental lithosphere in Middle Precambrian time. Furthermore, the dyke swarms extend for many hundreds of kilometres into the shield. They may thus constitute potentially sensitive recorders of how far thermochemical and tectonic disturbances associated with development of an active continental margin propagate into the continental interior. A preliminary step in determining the extent of these effects is to test the consistency of paleomagnetic, structural, and geochemical data *throughout* a given swarm to see if any longitudinal variations in these data exist that might be interpretable in terms of proximity to major rifting or collisonal events.

The purpose of this paper is to fulfil these objectives by using as a test case the 2.1 Ga old Kenora–Kabetogama dyke swarm, which covers an area of more than 10^5 km^2 in northern Minnesota and adjacent parts of Ontario and Manitoba.

Geology of the Kenora–Kabetogama dyke swarm

Structure and tectonics

The Kenora–Kabetogama (KK) dyke swarm trends in a northwesterly direction and has a longitudinal extent of more than 300 km (Fig. 1). At its northern end, the dykes disappear along a west-southwest-trending line about 10 km north of Kenora, although one thin (1.5 m) dyke terminating to the northwest occurs 50 km farther northwest (locality 21 in Fig. 1). At the south end of the swarm, geological and aeromagnetic evidence (Southwick and Day 1983; Chandler *et al.* 1984) indicates that the dykes pass underneath the basal sedimentary units of the Animikie Basin.

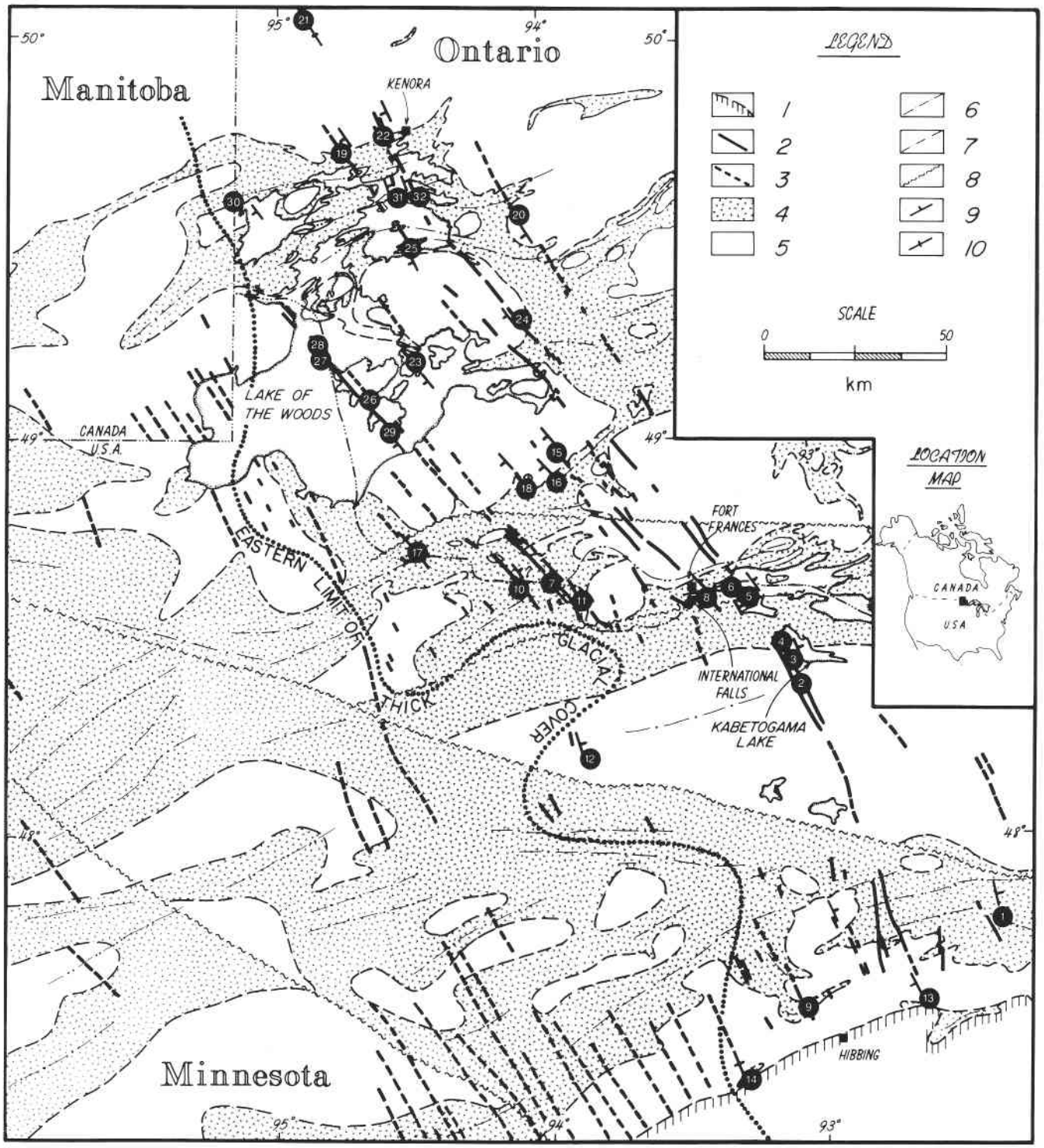


FIG. 1. Geological map showing the location of paleomagnetic sampling sites. Legend: (1) post-Archean cover rocks of Animikie Series; (2) dykes defined by outcrop and aeromagnetic anomalies; (3) dykes defined only by aeromagnetic anomalies; (4) mafic metavolcanics, metasedimentary rocks, and mafic gneiss; (5) predominantly granite or granite gneiss; (6) aeromagnetic lineaments; (7) fold axes; (8) fault; (9, 10) dip of dyke, inclined, vertical. In Minnesota most of the rocks west of longitude 93°30′W are covered by glacial drift. There the distribution of metavolcanics and dykes is based on aeromagnetic data. High-resolution aeromagnetic surveys (Chandler *et al.* 1984) extend south from International Falls to west of Hibbing where coverage roughly corresponds to the region of abundant dykes. The disappearance of these dykes to the north corresponds to a change from high- to low-resolution surveys.

Toward the west the swarm becomes covered by a thick mantle of glacial till (Fig. 1), and the full extent of the swarm in this direction can only be obtained from aeromagnetic surveys. High-resolution data from Minnesota (flying height 150 m, line spacing 400 m) (Chandler *et al.* 1984; V. W. Chandler, personal communication, 1984) show that the swarm extends in a southwest direction for at least 100 km beneath the till cover. The full width of the swarm is thus at least 300 km.

The KK swarm contains hundreds of dyke segments, some of which can be followed with the aid of aeromagnetic data for more than 200 km (Fig. 1). Dyke thickness ranges from 120 m to a few centimetres, although it is commonly between 20 and 70 m. The thinnest dykes do not occur in isolation but usually appear within several tens of metres of a comparatively thick dyke and may represent offshoots or apophyses. Dyke swarm trend in Fig. 1 varies from 150° in the south to 140° in the north, with changes in trend often occurring across major east–west trending dextral shears of probable Archean age (Southwick and Day 1983). Dyke dips are within 10° of the vertical, but the dominant dip direction changes from east to west in a northerly direction along the swarm (Fig. 1). In comparison with dyke swarms found elsewhere in the Canadian Shield (e.g., Ernst and Halls 1984), the steep dip ($\geq 80°$) of the KK dykes suggests they have suffered little or no tilting normal to dyke strike. The maximum amount of tilt parallel to strike, as given by the dip of the overlying Animikie strata at the southern end of the swarm, is less than 10°SE.

The KK swarm has been described by Southwick and Day (1983) as a "variety of failed arm" produced along preexisting fractures as a consequence of rifting along a continental margin presently buried beneath rocks of the Animikie Basin. No volcanics have yet been connected with the main rifting episode, although some of as yet undetermined age and petrology are known in the Mille Lacs Group within the Animikie Basin of Minnesota (Morey 1983) and occupy a stratigraphic position comparable to that of the dykes (Southwick and Day 1983). Large areas of rift-related volcanics (e.g., the Hemlock Formation, which is at least 1.8 Ga old) also occur along the Michigan–Wisconsin border about 250 km southeast of the swarm (Beck 1984).

Petrology and geochemistry

The dykes of the KK swarm form two distinct geochemical populations, one having high Mg and low Ti ($MgO > 7$, $TiO_2 < 1$ wt.%), the other low Mg and high Ti ($MgO < 5.5$, $TiO_2 > 2$ wt.%). This discovery, by Manzer (1978), was based on 54 whole-rock analyses, mostly on chilled margins and all from the southern end of the swarm in Minnesota. This result has been confirmed for the full length of the swarm by an additional 102 major-element analyses and 30 analyses of six minor elements obtained mostly from chilled margin samples collected at 30 sites (all except numbers 27 and 31 in Fig. 1) representing 13 different dykes (H. C. Halls, unpublished data). A similar but more extreme bimodality, in which the high Mg composition is more komatiitic, is a common feature of other ≈ 2 Ga old dyke swarms in the Canadian Superior Province (Fahrig *et al.* 1984). In the KK swarm both chemical types are quartz tholeiites and difficult to distinguish without the chemical data.

Petrographically, chilled margins are plagioclase- and clinopyroxene-phyric; the groundmass consists of finely felted plagioclase laths, granular clinopyroxene, granular opaques, and a complex, very fine-grained, secondary aggregate of opaque and green phyllosilicate minerals that has replaced interstitial glass. Primary phenocrysts of magnetite and ilmenite occur in some chilled-margin samples. Olivine was found only in the chilled margin of a dyke at locality 16 (Fig. 1), where it forms small phenocrysts that are mantled and embayed by rims of magnetite.

Dyke interiors range from diabasic to gabbroic in texture and typically contain magmatic hornblende and biotite in addition to clinopyroxene and plagioclase. Some dykes are quartz bearing and commonly display myrmekitic intergrowths of interestitial quartz and plagioclase (Southwick and Day 1983). The oxides of dyke interiors include blocky to bladed crystals of magmatic magnetite and ilmenite and also fine-grained secondary oxides related to a rather ubiquitous hydrous alteration of ferromagnesian silicates.

The hydrous alteration has affected all dyke rocks to varying degrees. This phenomenon has been discussed in detail for the Minnesota end of the swarm by Hanson and Malhotra (1971), who attributed the alteration to burial metamorphism, and by Southwick and Day (1983), who favoured alteration by residual or late magmatic fluids. The severity of hydrous alteration varies sharply over distances of several centimetres to metres at a given sample site but appears to decrease gradually but rather irregularly in a northwest direction along the length of the swarm. The alteration involves the production of Mg-rich amphibole and was thought (Southwick and Day 1983) to be at least partly responsible for the two compositional types. Although the alteration may locally enhance MgO content, the new geochemical data show convincingly that the two chemically distinct compositions are primary features, because each characterizes individual dykes along strike lengths ranging from 50 to over 300 km. The two compositional types have not been found together in any composite dyke, so their relative age cannot be determined on field evidence.

The oxide mineralogy of the dyke rocks clearly is affected by the alteration process. In chilled margins there is petrographic evidence of secondary oxide production as primary ferromagnesian phenocrysts were altered to chlorite, sericite, clinozoisite, and opaque phases and also of oxide destruction as granular Ti–Fe oxides in the groundmass were altered to leucoxene and granular sphene. Dyke interiors exhibit extensive secondary oxide development upon hydrous alteration of clinopyroxene (and biotite to a lesser degree) and also show substantial evidence of ilmenite converting to leucoxene. In general, hydrous alteration is greater in dyke interiors than in margins. Although the compositions of the titanomagnetites are not known in detail, Curie point determinations on 10 chilled-margin samples all indicate the presence of a Ti-poor or almost pure magnetite.

Dyke-parallel layering occurs at sites 3, 4, 6, 14, 17, 22, 28, and 29. It may be developed at other sites where dyke thickness exceeds about 20 m, but because of absence of smooth horizontal outcrops this could not be verified. At site 28, erosional structures are observed, similar to that reported by Williams (1976) in a Greenland dyke where layers of normal diabase alternate with more felspathic diabase. Channel structures defined by these layers give the appearance of trough cross-bedding. The channels are asymmetric in horizontal section, with the southeast sides making a smaller angle with the dyke margin. If these channels are symmetric in cross section normal to their axes, the field observations suggest that their axes dip steeply to the southeast. A dominantly vertical

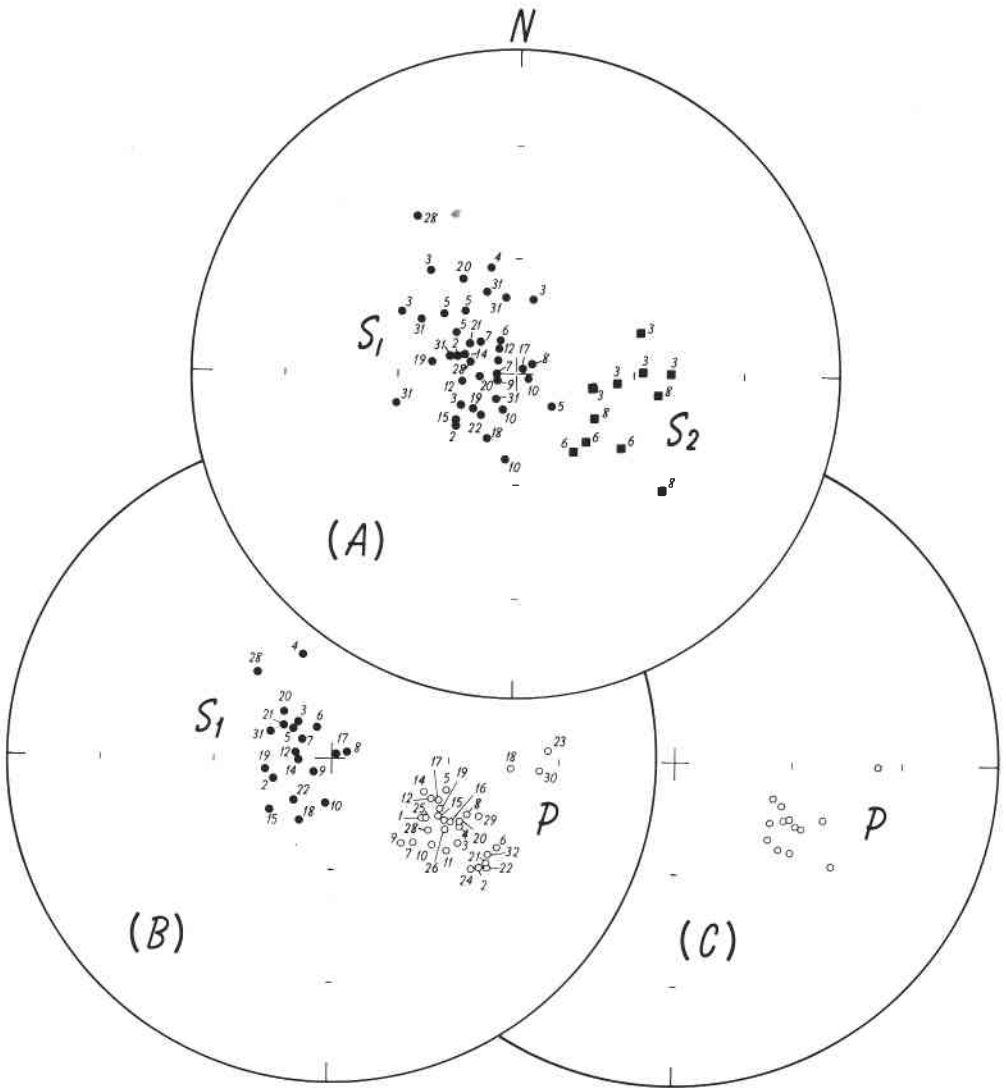


FIG. 2. Equal-area steronets: (*A*) all samples containing secondary S_1 and S_2 components; (*B*) site directions for S_1 and primary P component; and (*C*) dyke-averaged directions for P component. Open (closed) symbols are upward- (downward)-pointing remanence directions.

magma trajectory with a slight northward component of flow may thus have characterized dyke emplacement during formation of the channels.

The best available estimate of the age of the KK swarm is that by Beck and Murthy (1982), who obtained a Rb−Sr isochron of 2120 ± 67 Ma on a composite dyke of the low-Mg variety at site 3 (Fig. 1). Available K−Ar dating (Hanson and Malhotra 1971) allows the possibility for the more Mg-rich dykes to be as old as 2.2 Ga.

Geological field relations (Southwick and Day 1983) and aeromagnetic data (Chandler *et al.* 1984) show that the dykes pass under, and are thus older than, the basal formation of the Animikie Group, which was deformed by the Penokean Orogeny about 1.85 Ga ago (Sims *et al.* 1980).

Paleomagnetic investigations

Experimental procedure

A total of 31 paleomagnetic sites were studied along the length and breadth of the exposed part of the swarm (Fig. 1). Five to 10 large hand samples, representative of chilled margin and interior and oriented using both magnetic and sun compasses, were collected at most sites. About half the sites included samples of adjacent country rock, for baked-contact tests (Everitt and Clegg 1962).

Of the several core specimens drilled from each sample, at least one was subjected to detailed alternating field (AF) demagnetization using a Schonstedt single-axis instrument. For 50 samples, another specimen was thermally demagnetized using a Schonstedt TSD-1 instrument. After each demagnetization step, the direction and intensity of remanent magnetization were measured either on a doubly shielded DIGICO or a Schonstedt spinner magnetometer.[1]

[1]A magnetic field due to concentration of flux lines occurs at the top of the DIGICO shield, where samples are loaded. This field can be sufficiently strong to generate a significant viscous remanent magnetization (VRM) during loading in samples in advanced stages of thermal or AF demagnetization. Because of the DIGICO spin sequence, this VRM ultimately appears as a component with shallow inclination and directed about 330° with respect to the sample arrow. To avoid the appearance of the unwanted component, Helmholtz coils are used to cancel the field at the top of the DIGICO shield.

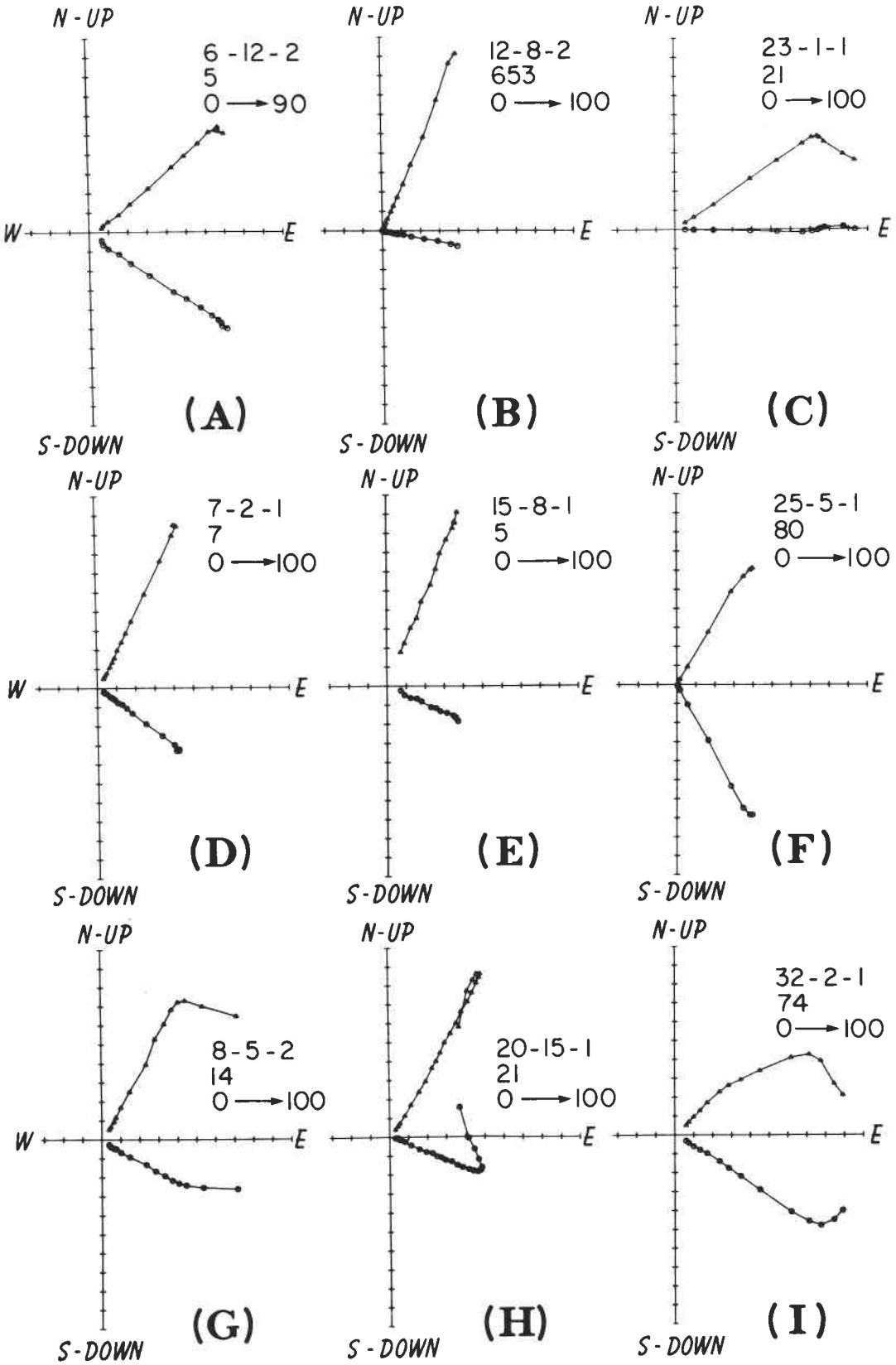


FIG. 3. Vector diagrams showing decay of remanence component P, interpreted to have formed during initial dyke cooling. Circles and triangles represent directions projected onto the horizontal and vertical planes, respectively. Numbers shown sequentially on each plot are the sample number, scale division (10^{-3} A/m), and range of alternating field treatment (mT). Distances (cm) of samples from dyke chilled margin (all in dyke except for E, which is in granite host rock): (*A*) 3×10^2, (*B*) 9, (*C*) 30, (*D*) 10, (*E*) 20, (*F*) 1, (*G*) 9, (*H*) 23, (*I*) 15. Dyke widths are given in Table 1.

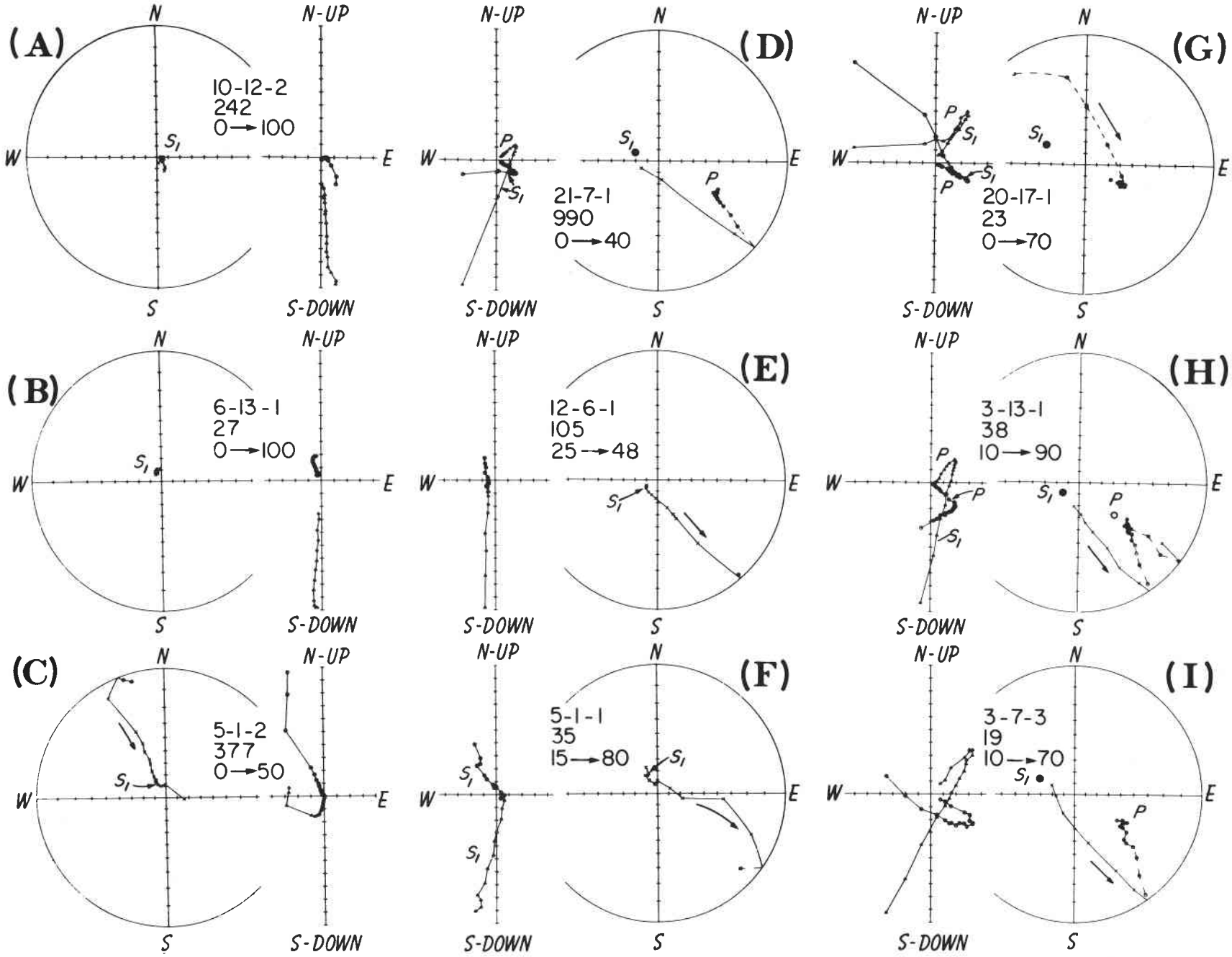


FIG. 4. Vector diagram — stereonet pairs of dyke samples showing an interpreted secondary component S_1. The degree of S_1 overprinting on P generally decreases from left to right across the diagram. Symbols and notations as in Figs. 2 and 3. Distances (cm) of samples from chilled margin: (*A*) 3×10^2, (*B*) 3×10^3, (*C*) 2, (*D*) 1.5, (*E*) 2.4×10^2, (*F*) 2, (*G*) 10^3, (*H*) 2×10^3, (*I*) 1.8×10^3.

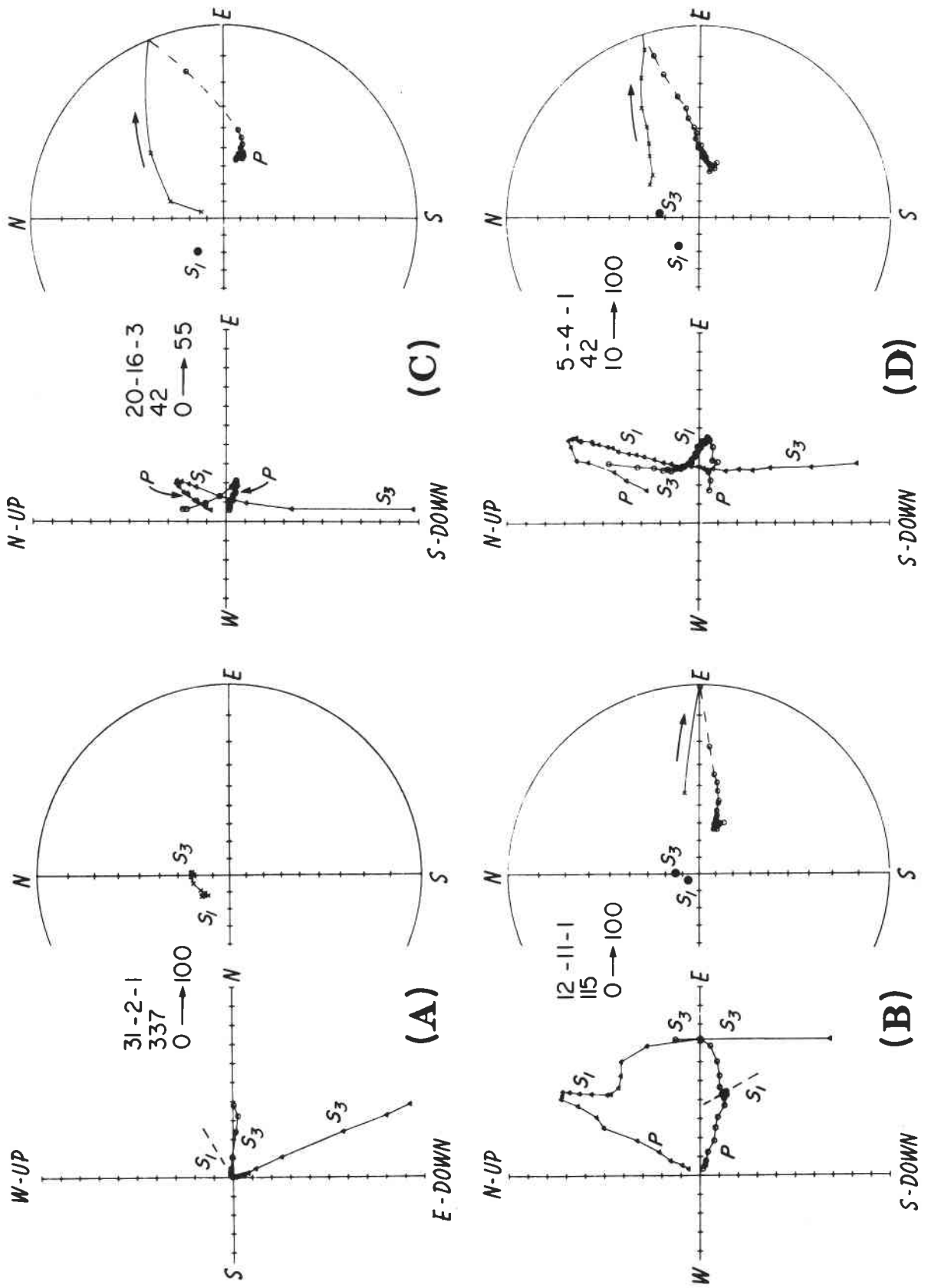


FIG. 5. Vector diagram — stereonet pairs of dyke samples showing secondary component S_1 in association with another component S_3 interpreted to be a VRM acquired in the present Earth's field. In all cases S_3 has a lower coercivity spectrum than S_1. Symbols and notations as in Fig. 4. Distances (cm) of samples from chilled margin: (*A*) 1, (*B*) 8, (*C*) 12, (*D*) 1.5×10^3.

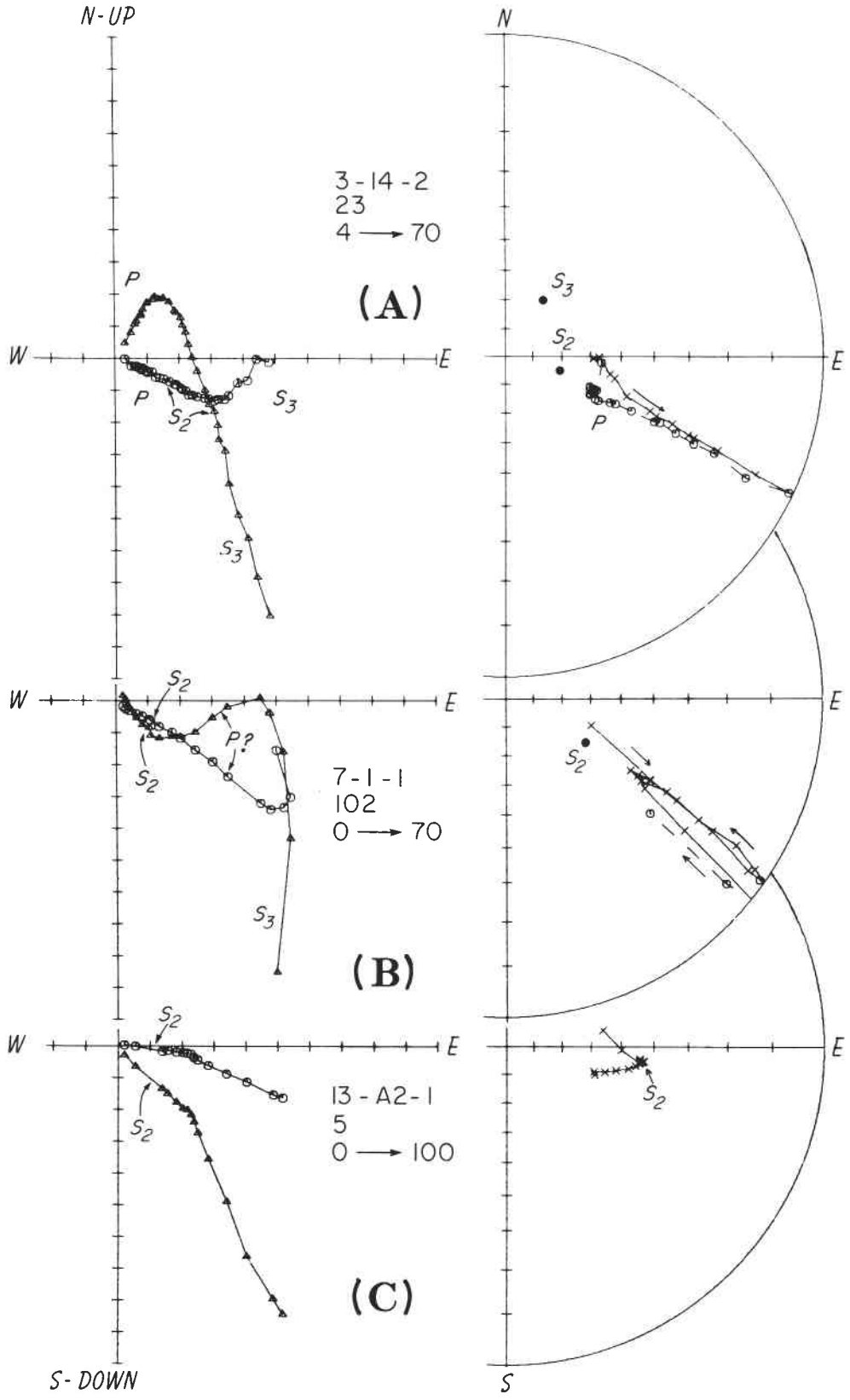


FIG. 6. Vector diagram – stereonet pairs of three dyke samples showing a secondary component S_2. Symbols and notations as in Fig. 4. Distances (cm) of samples from chilled margin: (*A*) 12, (*B*) 5, (*C*) 60. It is possible that in (*A*) and (*B*) S_2 represents a contaminated S_1. Note that (*B*) gives an estimate of P that is much too shallow, indicating there may be P and S_1 components whose coercivity spectra are very similar over two intervals that generate the S_2 and shallow P components. S_2 is similar to S_1 as it also occupies a coercivity range intermediate between those of S_3 and P.

TABLE 1. Summary of paleomagnetic results for the interpreted primary magnetization component from the Kenora–Kabetogama dyke swarm[a]

Site	Lat. (°)	Long. (°)	Trend (°)	Dip (°)	Width (m)	N	N_S	N_R	J	D (°)	I (°)	D' (°)	I' (°)	k	α_{95} (°)		N	$\bar{D}'$ (°)	$\bar{I}'$ (°)	k	α_{95} (°)
20	49.56	94.10	170	85W	50	11	8	0	10	115.6	−53.7	115.4	−53.9	110	5.2						
5	48.62	93.23	135	—	30	6	1	2	15	105.9	−59.2	104.3	−60.1	290	15.5						
4	48.45	93.08	190	85E	60	4	1	2	20	117.8	−52.2	116.3	−53.5	54	36	→	3	116.6	-54.2	330	6.8
3[b]	48.42	93.07	135	90	70	13	7	3	20	124.0	−50.0	122.7	−51.5	82	6.0						
21	50.08	94.88	145	90	1.5	9	6	0	300	122.7	−43.0	123.2	−42.4	100	6.6						
22	49.77	94.60	170	80W	70	6	3	0	10	123.7	−41.6	123.9	−41.4	230	8.2						
32	49.64	94.46	160	85E	70	5(1)	4	0	5	120.5	−42.9	120.5	−42.9	117	6.9						
24	49.37	94.16	120	80W	70	11	3	7	25	126.7	−44.0	126.5	−44.4	180	5.5	→	6	122.7	-42.7	830	2.3
6	48.63	93.25	132	85E	>40	8(1)	2	5	10	118.0	−40.6	116.9	−42.1	26	17.8						
2	48.37	93.00	163	85W	>10	6	3	3	15	126.9	−41.4	125.6	−43.4	160	6.8						
19	49.71	94.70	135	80W	25	7(2)	8	0	35	113.1	−59.6	113.3	−59.4	160	4.5						
25	49.48	94.48	135	80W	≈50	5	3	0	15	120.7	−62.1	120.5	−62.1	70	14.8						
8	48.62	93.37	155	85E	50	8(2)	10	0	5	112.7	−51.6	111.5	−52.7	177	4.4	→	4	115.5	-58.0	320	5.1
13	47.60	92.56	170	80W	10	7	5	0	15	120.0[c]	−55.0	117.7	−57.5	98	7.8						
30	49.58	95.15	135	83W	40	7	3	1	10	92.0	−37.0	92.6	−36.2	280	6.2						
23	49.20	94.30	135	80W	20	2	2	0	20	88.0	−33.9	87.8	−33.9	1000	7.9	→	3	90.9	-38.3	180	9.2
18	48.88	94.10	135	80W	20	7	4	1	10	93.3	−44.4	92.7	−44.6	86	9.0						
28	49.23	94.82	130	90	>25	8	6	0	15	124.6	−59.9	124.4	−59.9	76	7.7						
26	49.11	94.67	125	—	80	5	5	0	15	120.8	−55.8	120.6	−55.9	440	3.7						
7	48.64	93.90	130	75E	50	7(6)	10	2	20	134.5	−60.4	133.6	−61.2	530	2.0	→	4	128.7	-60.1	260	5.7
9	47.58	93.00	150	—	40	11	5	0	40	140.0[c]	−61.0	138.3	−62.8	41	12.1						
12	48.18	93.80	150	85E	40	10(1)	7	3	15	112.2	−62.1	110.5	−62.8	180	3.9						
14	47.37	93.22	155	90	20	5(1)	4	1	60	111.0[c]	−64.0	108.1	−65.1	34	14.5	→	2	109.4	-64.0	2100	5.5
1	47.76	92.25	175	76E	7	3	2	1	5	123.0[c]	−60.0	120.5	−62.4	380	8.2						
15	48.98	93.91	140	87W	>30	7(2)	4	3	10	117.6	−58.5	116.8	−59.0	220	4.8						
16	48 90	98.91	120	85W	30	6	6	0	10	117.4	−55.8	116.6	−56.4	510	3.0						
11	48.60	93.82	170	85E	50	6(1)	4	3	20	128.4	−52.8	127.5	−53.8	60	9.3						
10	48.65	94.05	167	85E	50	12(1)	8	0	15	129.7	−55.7	128.9	−56.5	40	8.9						
17	48.70	94.45	135	85E	70	9	6	1	15	111.4	−61.2	110.6	−61.4	180	4.7						
29	49.03	94.54	125	85W	>50	8	3	2	20	110.3	−49.9	110.1	−50.0	230	6.9						

No. of dykes = 13
Mean direction = 115.9°, −55.6
k = 68; α_{95} = 5.1°
Pole lat. = 42.7°S
Pole long. = 4.2°E
dm = 7.3°, dp = 5.2°

NOTES: N = number of dyke samples or sites measured; numbers in parentheses indicate the number of country-rock samples that contributed data to the site estimate; N_S = number of samples (dyke + host) yielding stable end points used in site estimate; N_R = number of samples (dyke + host) yielding remagnetization circles used in site estimate; J is approximate natural remanent magnetization (NRM) intensity to nearest 5×10^{-1} A/m; D, I = measured declination, inclination at site; D', I' = directions converted to a common site at Kenora (lat. 49.8°N, long. 94.5°W); k, α_{95}, dp, and dm are the precision parameter, radius of 95% confidence circle about a mean direction, and semi-axes of 95% confidence ellipse about the pole, respectively. Columns 4–6 give the dyke parameters at each site.

[a]Site 27 is omitted because it was severely lightening struck; site 31 is not included because it yielded only the S_1 component (Table 2).

[b]No difference in paleomagnetic direction was found between a central dioritic dyke and an older diabase one. This is the site used by Beck and Murthy (1982) to obtain their Rb–Sr isochron; it is described in more detail by Southwick and Day (1983).

[c]A structural correction for strike = 70°, dip = 7°SE as given by overlying and nearby Animikie strata has been applied to these sites.

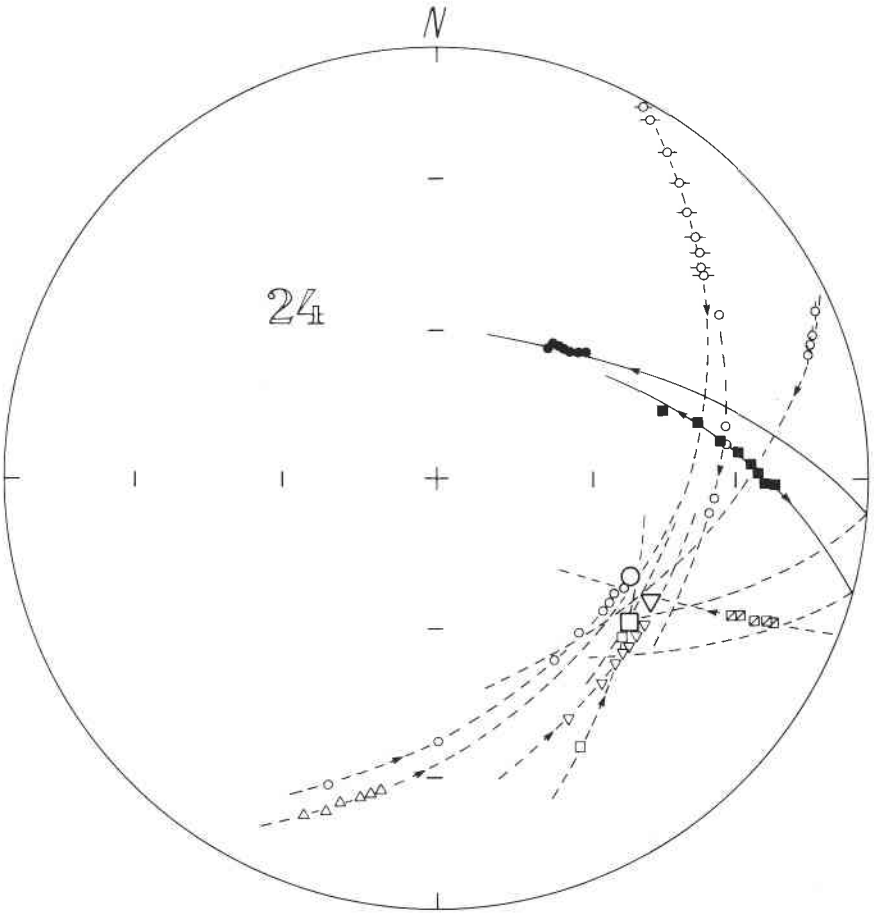


FIG. 7. An example of the use of remagnetization circles to define the P component for site 24. A strong, highly scattered, low-coercivity secondary component generates the wide azimuthal spread of the circles. The large symbols are stable end points and are defined for only 3 of the 10 samples.

In AF demagnetization (which was the most effective means for component separation) field increments were variable and adjusted so that no more than five degrees of vector movement occurred after each demagnetization step. Demagnetization was continued as long as a smooth change in intensity and direction was maintained.

At the first onset of erratic behaviour, the run was either terminated or, if further data were required to better define a removed component, the run was continued using the following procedure. In single-axis demagnetizing instruments it is found, particularly when residual intensity is less than 20% of the initial value, that the direction and intensity of the remanence vector depend upon the last axis of the specimen that is demagnetized. For a given AF value, the specimen is demagnetized sequentially along its x, y, and z axes before measurement. The specimen is then again demagnetized along the x and y axes, followed in each case by a measurement. The resultant of the three measurements is then taken to represent the remanence vector for the AF value. Using this empirical method it is often possible to continue trends in remanence behaviour that have been defined at lower AF values.

Remanence components were separated using the methods of principal component analysis (PCA) (Kirschvink 1980) and remagnetization circles (Halls 1976). Kirschvink's PCA computer programme was modified to generate all possible linear segments comprising three or more consecutive points and having a mean angular deviation (MAD) $\leq 5°$. Segments chosen to represent remanence components were selected on the basis of lowest MAD value and if their directions were not part of a progressive directional change when compared with those of adjacent and overlapping segments. All components obtained from PCA were checked for validity on vector diagrams and stereonet plots before final acceptance.

Many samples were dominated by a low-coercivity component, often having a variable direction within a site. Samples showing this behaviour frequently failed to attain stable end points during AF treatment but instead followed remagnetization circles. The points defining the remagnetization circles were fitted by a least-squares method (Halls 1976) and accepted if the mean radius of the 95% confidence ellipse about the pole (normal to the plane of the circle) was $\leq 5°$. The pole data and any stable end points obtained from independent samples in the same site were then combined to give a site direction using the method of Bailey and Halls (1984).

Component identification

Three remanence components, designated P, S_1, and S_2, have been identified in KK dykes, and their directions are shown in Fig. 2.

The P component is the most common, as it is found in both dyke interior and margin samples of every site except one (No. 31). When occurring with other components it *always* resides in the highest coercivity remanence carriers and thus can be isolated as a stable end point by AF demagnetization. It

TABLE 2. Summary of paleomagnetic results for the secondary component S_1

Site	N	D' (°)	I' (°)	k	α_{95} (°)
2	2	254.5	74.9	46	37
3	5	320.0	77.0	11	24
4	2	346.3	62.5	390	13
5	4	310.7	77.5	24	19
6	1	333.9	80.4		
7	2	307.0	81.3	300	15
8	1	60.0	86.0		
9	1	249.9	85.2		
10	3	186.0	79.9	55	17
12	2	284.6	80.9	94	26
14	2	286 8	80.8	150	20
15	1	231.5	70.6		
17	1	55.2	88.6		
18	1	208.2	73.4		
19	2	260.8	74.0	57	34
20	2	314.6	74.0	22	56
21	1	304.5	76.5		
22	1	220.0	77.0		
28	2	318.0	61.6	10	99
31	6	296.9	73.9	25	14

No. of sites = 20
Mean direction = 285.4°, 81.2°
k = 47, α_{95} = 4.8°
Pole lat. = 51°N
Pole long. = 120.6°W
dm = 17, dp = 16

NOTES: Many northerly sites (lower half of the table) have only one overprinted sample, and others show poor directional agreement between samples. By contrast, the results from southerly sites (upper half of table) show better within-site agreement and are based on more samples per site. Also only two southern sites (1 and 13), compared to eight northern ones, were not noticeably overprinted by S_1.

is best defined in chilled margins, where it often occurs to the exclusion of other components; samples within about 25 cm of the margin have the highest median destructive fields (20−30 mT) and give clear linear decays to the origin in vector diagrams (Fig. 3).

In dyke interiors, median destructive fields are closer to 10 mT, and the P component, if it exists, is dominated by one or more low-coercivity (<25 mT) remanences comprising S_1, S_2, a present Earth's field (PEF) component designated S_3, and other more random components. Despite the relatively large size of the S components, it is still possible in many samples to obtain the direction of the P component. Several samples of P separation from S are given in Figs. 4−6.

A summary of paleomagnetic site data for the P component is given in Table 1. Some sites, which are believed to be from the same dyke, have been subsequently combined to give directions for individual dykes. The evidence used to correlate different dyke segments will be discussed later.

Component S_1 is found in about two thirds of the sites, but often in only one or two samples per site can its direction be determined with the required degree of reliability. It occurs either with P or by itself (Fig. 4). When found alone, it is generally confined to dyke interiors. The direction of S_1 (≈285°, +81°, Table 2 and Fig. 2) is close to that of the present Earth's field (355°, +76°) but does *not* represent a recently acquired VRM (S_3) component, because *both* S_1 and a lower coercivity S_3 component occur together in several samples (Fig. 5).

Component S_2 is less common than S_1; it is found in only five sites and has a direction of about 110°, +60° (Fig. 2*A*). It appears to occupy a coercivity niche similar to that of S_1, as it is intermediate between P and S_3 (Fig. 6) but is never found together with S_1 in the same specimen.

In the lowest coercivity ranges (below 15 mT), components are often found that lack directional consistency within a site. These components, which may represent laboratory VRM's or lightening-induced remanences, do not appear to have paleomagnetic significance. Even where they dominate the magnetic signature at a site, they can sometimes be used, as in site 24, to obtain an estimate of the P direction via converging remagnetization circles (Fig. 7). For such sites it is essential to define in detail the low-coercivity part of each sample demagnetization path.

Origin of components

Component P is interpreted to be of primary origin for two reasons:

(1) In Fig. 2*B* site-mean directions for the P component fall into three clusters: one formed by sites 2, 6, 21, 22, 24, and 32; a second by sites 18, 23 and 30; and the third, larger cluster by all remaining sites. The first cluster is particularly noteworthy because it is made up of sites that are known or suspected on the basis of projected strike extension and aeromagnetic anomaly correlation to be part of the same dyke (Fig. 1). On the basis of strike extension, the second, smaller cluster could also be from a single dyke.

Support for these correlations and for others that are less obvious on the sole basis of paleomagnetism may be obtained from the whole-rock geochemical data. A first separation of dykes can be made using the two different magma types, distinguished on the basis of their MgO and TiO_2 contents. However, there is a sufficient difference in MgO, P_2O_5, Zr, and Cr *between* individual dykes *within* each compositional type to permit along-dyke correlations. The geochemical results for MgO and P_2O_5 (Fig. 8 and Table 3) show clear correlations for individual dykes that agree well with correlations made on the basis of the P component data.

Because the direction of P appears to characterize a particular dyke along strike lengths exceeding, in one instance, 300 km, it is undoubtedly related to its time of intrusion. A secondary component of such regional extent might be expected to affect all the dykes and thus destroy any between-dyke variation in paleomagnetic direction.

(2) A number of baked-contact tests on samples of host rock adjacent to dyke margins (within a few centimetres) yield directions similar to those for P (Fig. 9), whereas those farther away generally give erratic results. The tests are not fully positive because it cannot be shown that the host rocks carry a pre-dyke remanence. However, the results are consistent with preservation of the host-rock component at the time of baking and therefore with P being primary in origin. Although there is close agreement between dyke and host-rock directions at sites 7, 8, 15, and 19, other sites such as 12, 18, 20, and 32 show host-rock directions with comparatively low declinations (Fig. 9). The host rocks were either metavolcanics or gneisses with a steep, east- to east-northeast trending foliation, inviting speculation that a magnetic anistotropy is responsible for the anomalous remanence directions. Measurements of anisotropy

TABLE 3. Selected mean sample major-element oxide and trace-element analyses for sites that on the basis of their geochemistry belong to the same dyke

	Dyke	Site	N	N'	MgO (wt.%)	SD	TiO_2 (wt.%)	SD	P_2O_5 (wt.%)	SD	MgO / MgO + FeO	SD	Zr (ppm)	Y (ppm)	Ni (ppm)	Cr (ppm)
NE	1	20	4	4	4.87	0.24	2.12	0.08	0.210	0.008	0.235	0.008	128	51	63	60
		3–5	6	6	4.96	0.10	2.13	0.02	0.200	0.005	0.232	0.005	132	45	63	60
	2	21	5	5	5.03	0.27	2.21	0.07	0.217	0.004	0.245	0.009	143	42	69	109
		22	1	1	5.27	—	2.19	—	0.21	—	0.259	—	141	36	73	99
		32	3	1	5.58	—	2.23	—	0.21	—	0.267	—	143	43	70	130
		24	7	7	5.54	0.08	2.15	0.03	0.208	0.008	0.264	0.003	142	44	92	111
		6	3	3	5.78	0.10	2.24	0 09	0.207	0.005	0.273	0.003	141	37	63	134
		2[a]	3	1	5.87	—	2.19	—	0.20	—	0.276	—	135	43	57	126
	3	19	3	3	7.76	0.21	0.91	0.03	0.07	0.0	0.433	0.014	57	22	87	211
		25	4	3	7.86	0.03	0.93	0 00	0.07	0.0	0.413	0.000	45	22	110	214
		8[a]	5	2	7.56	0.37	0.92	0.03	0.07	0.0	0.413	0.001	59	19	120	192
	4	30	4	4	7.77	0.32	0.73	0.04	0.06	0.0	0.428	0.010	50	17	100	178
		23	2	2	7.94	0.07	0.69	0.04	0.06	0.0	0.425	0.006	45	15	110	236
		18	2	2	8.06	0.11	0.67	0.03	0.06	0.0	0.439	0.001	45	12	74	252
	5	28	3	3	5.46	0.24	2.15	0.14	0.193	0.016	0.250	0.014	131	39	46	90
		26	3	3	5.49	0.28	2.04	0.07	0.180	0.010	0.261	0.010	128	41	60	92
		7	3	2	5.52	0.14	2.04	0.07	0.175	0.007	0.260	0.008	123	38	62	97
		9[a]	3	3	6.09	0.17	1.91	0.10	0.170	0.010	0.291	0.010	130	38	47	96
SW																

NOTES: Major-element oxides were done by X-ray fluorescence (XRF) whole-rock analysis (estimated uncertainty about 5%), the Zr, Y, and Cr were done by XRF on pressed pellet (uncertainty ±2–3 ppm), and the Ni was obtained by plasma emission spectroscopy (uncertainty ±5%). All analyses and uncertainty quotations were supplied by X-Ray Assay Laboratories, Don Mills, Ontario. Although the third decimal figure is not significant and is thus not quoted for individual sample analyses, it *is* quoted for the mean sample values where necessary to illustrate small longitudinal trends. No statistical analysis has been done to determine whether these trends are meaningful. All samples were collected less than 2% of dyke width away from the chilled margin. Of the total number of samples measured (N), results were only included for those samples (N') where the loss on ignition (LOI) was less than 2.0% by weight. All samples were 2.5 cm × 2.5 cm paleomagnetic cores collected from different hand samples. For the oxide analyses, SD is the standard deviation. Trace-element analyses were performed on single-site composites comprising equal aliquots of each sample from that site.

[a]In these sites there is a positive correlation between MgO and LOI. In sites 2 and 8 the majority of samples had LOI > 2.0.

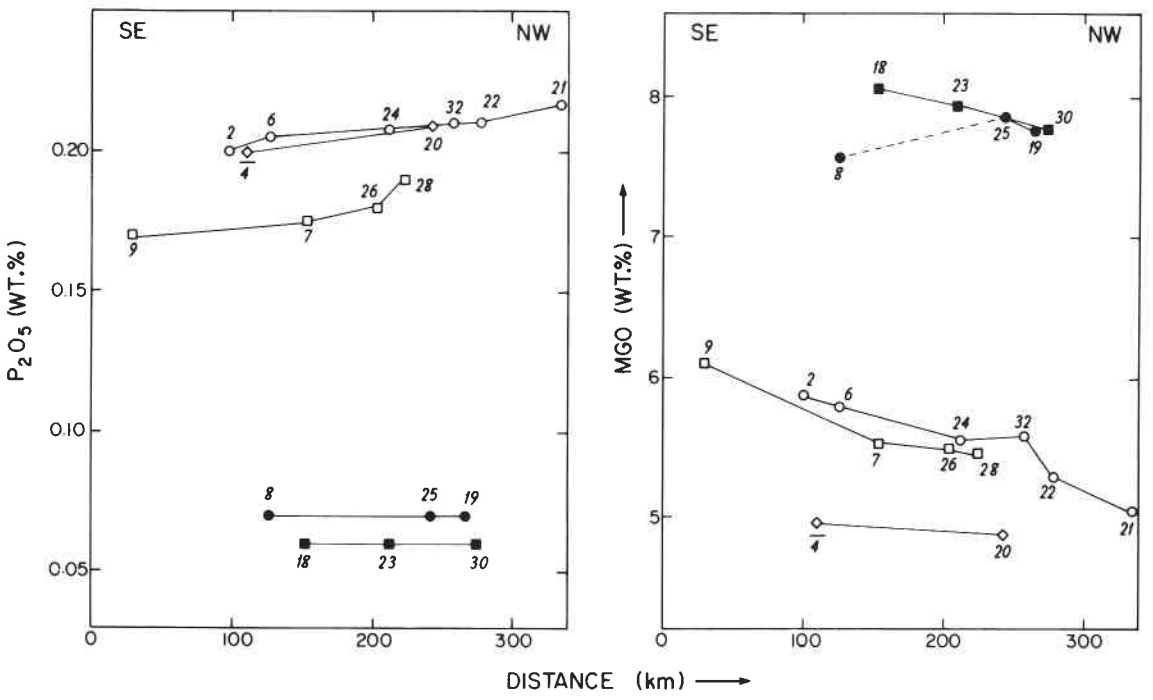


FIG. 8. Longitudinal geochemical profiles for MgO and P_2O_5 in five dykes. Individual site results (see Table 3) are means of several chilled-margin samples. The distance is measured northward from the line along which the dykes emerge from beneath the Animikie Basin.

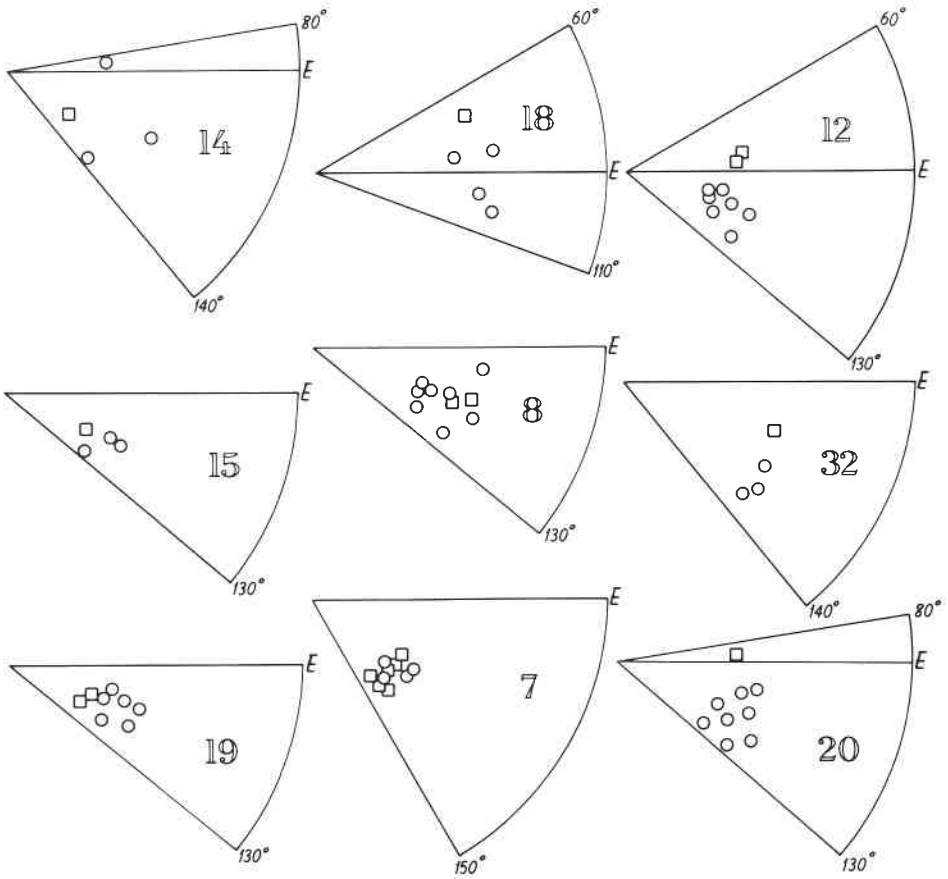


FIG. 9. Equal-area stereonets for nine paleomagnetic sites comparing primary component directions for dyke samples (○) with those from adjacent host-rock samples (□).

of magnetic susceptibility (AMS) were thus made on several dyke and host-rock samples from the eight sites, using a Sapphire SI-1 instrument. AMS values given by $(k_{max} - k_{min})/k_{min}$ averaged about 3% for dyke rocks, and for well-behaved and anomalous sites mean host-rock values were about 10 and 20%, respectively. Host-rock susceptibility ellipsoids were oblate, with major and intermediate axes lying within the foliation plane. The AMS results therefore support the idea that at the anomalous sites the fabric-induced magnetic anisotropy has been sufficient to deflect the host-rock magnetization from the paleofield direction (as given by the dyke remanence) towards the foliation plane.

If the P component is primary, all S components are secondary. The secondary origin for S_1 and S_2 in terms of chemical remanent magnetization (CRM), partial thermal magnetization (PTRM), or a combination of the two (PTCRM) is suggested by the following observations.

(1) A comparison of thermal decay curves for single-component samples carrying either S_1 or P (Fig. 10) shows that whereas the P component has a square-shouldered decay typical of many TRM's, the S_1 curve shows more distributed unblocking temperatures, a common feature of low-temperature CRM or PTCRM remanences. S_1 has unlocking temperatures close to the Curie point of pure magnetite, which also appears to be the dominant carrier for the P component. Therefore, it is unlikely that S_1 could be a pure thermal overprint: otherwise it would have erased the P component.

(2) S_1 and S_2, although often occurring in the same site (e.g., No. 3), do not coexist in the same specimen, as they appear to occupy similar coercivity ranges. This relation suggests that each arises from rather similar geological causes. For example, two events having similar temperature–time behaviour would be capable of resetting the same grains.

(3) Development of either S_1 or S_2 tends to be less in dyke margins and more intense in interiors, where both grain size and hydrous alteration are greater. Neither component is conspicuous where the overall level of dyke alteration is low. For example, in the Lake of the Woods area, all dykes at sites 15–30 (excluding sites 25 and 29) are relatively fresh. Apart from sites 25, 29, 31, and 32, all the more altered sites are found in Minnesota (sites 1–6, 9–14). Table 2 and Fig. 2 show a high proportion of these sites exhibiting S_1 and (or) S_2 development.

The foregoing evidence is consistent with a mode of remanence acquisition for S_1 (and S_2, because of its similar nature) involving low-temperature heating (less than ~300°C) and chemical production of new oxide phases during hydrous alteration. An important problem, to be addressed in the next section, concerns the nature of the geological events that produced these remanences.

Discussion

Hydrous alteration in the KK swarm was considered by Southwick and Day (1983) to be primarily of deuteric origin because alteration increases towards dyke interiors and primary minerals indicate hydrous conditions in the late-stage magma. Although deuteric alteration should be complete in a

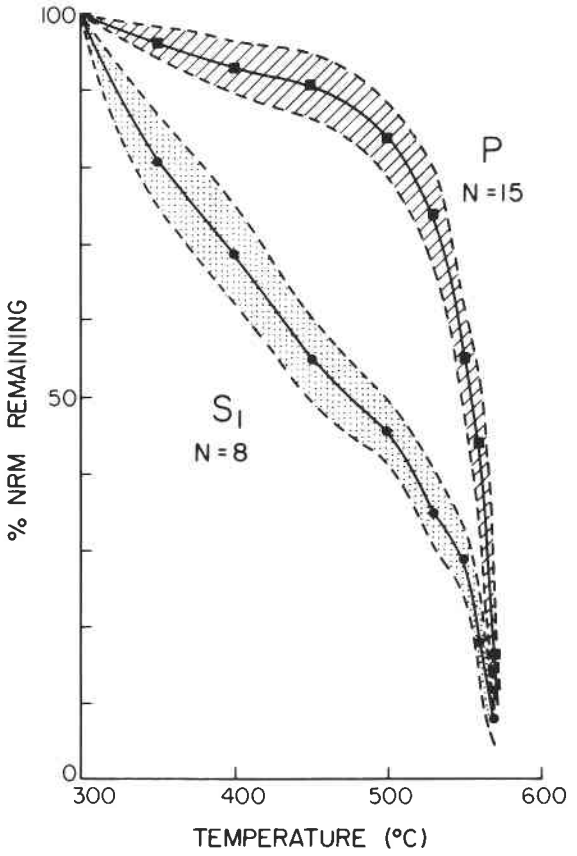


FIG. 10. Mean thermal demagnetization decay curves for samples carrying only the P or S_1 component. The normalization has been carried out with respect to the remanent intensity after 300°C heating, a temperature that corresponds to the final removal of a low unblocking-temperature component that is present in addition to P or S_1. The envelopes around the curves are at one standard deviation. It is clear that the S_1 component has remanence carriers with lower and more distributed unblocking temperatures.

few hundred years after dyke injection, the directions of both S_1 and S_2 are sufficiently different from P to suggest that they are much later than the deuteric activity. The reversed nature and the steeper inclination of S_1, compared with P for example, indicate an elapsed time after initial cooling of *at least* several millennia. An explanation for either S_1 or S_2 must therefore be sought in thermochemical changes that are later and not necessarily related to deuteric activity.

Locally intense alteration of chilled margins is found in some Minnesota sites (e.g., 13, 14, and 4) and may be associated with geological events occurring along the once active continental margin to the south. These events include rifting and magmatism, formation of the Animikie sedimentary sequence, and the Penokean Orogeny. Each one of these events may be capable of promoting hydrous alteration in adjacent regions of the continental interior, where raised levels of heat flow would generate more chemically active and vigorous groundwater circulation. The effectiveness of groundwater in contributing not only to deuteric but also to later hydrous alteration depends, in part, on the ease by which water is supplied to the dyke from adjacent host rocks. Where granite or granite gneiss forms the country rock, as in the southern Lake of the Woods area, KK dykes are relatively fresh, but where there are more highly deformed and sheared supracrustal sequences, as in the Kenora and Minnesota regions, the dykes are more altered.

It is therefore proposed that supracrustal sequences, having a higher fracture porosity and permeability, offer preferred routes for circulating groundwater. This water may be incorporated into the magma during dyke emplacement, thus enhancing deuteric alteration, but if channelways remain open, further alteration of the dyke may be expected during subsequent periods of active groundwater circulation. In this way the positive correlation between S development and hydrous alteration level might be explained. Because both S_1 and S_2 postdate deuteric alteration and are more conspicuously developed in southern sites, they may be coeval with hydrous alteration produced during different stages of continental margin development. The following arguments suggest that S_1 and S_2 are related to the initial rifting and final orogenic stages, respectively.

The S_1 pole position (Table 2) is virtually identical to that obtained by Dunlop (1983) from an interpreted primary remanence in a 150 km long, west-northwest-trending quartz tholeiite dyke from the Wabigoon area just 150 km east of the KK swarm. These pole positions in turn resemble those of the 2.15 ($\pm$0.25) Ga quartz tholeiite and 1.15 Ga olivine diabase dykes east of Lake Superior, which belong to the Preissac and Abitibi swarms, respectively (Card *et al.* 1981). There is therefore ambiguity in the S_1 age based on paleomagnetism alone. However, the quartz tholeiite composition of the Wabigoon dyke suggests a closer affinity with Preissac magmatism, and the proximity of this dyke to the KK swarm suggests that S_1 has a Preissac age. The radiometric ages of the KK and Preissac dykes are not significantly different, so S_1 can only be younger than P by at most a few tens of millions of years. The cause of the S_1 component is tentatively ascribed to increased crustal heat flow and groundwater circulation along the margin of the newly rifted continent.

Although the direction of S_2 is less well defined than that of S_1, its pole position plots approximately on a 1.8 Ga old segment of the Precambrian apparent polar wander path (see Schutts and Dunlop 1981) and may thus be related to the Penokean Orogeny.

On the basis of the P component direction (Table 1), the Kenora–Kabetogama swarm appears to be coeval with Early Proterozoic Richmond Gulf volcanics (and associated dykes) on the east shore of Hudson Bay (Schwarz 1976) and also with some members of the north-trending Marathon swarm of quartz tholeiite dykes north of Lake Superior (H. C. Halls, unpublished data). This swarm also contains dykes having S_1 type directions (Fahrig *et al.* 1965). Therefore, from west to east around the northern margin of Lake Superior, primary remanences of $\approx$2.0 Ga dyke swarms change from a P direction in the KK dykes to S_1 type directions for Preissac dykes, with the intervening region from Wabigoon to Marathon characterized by dykes having either P or S_1 directions. If the Preissac remanence *is* younger than P, a model can be proposed whereby major rifting centres propagated or moved eastwards along the southern margin of the Superior Province, spawning a series of failed arm dyke swarms in the process.

One final issue concerns longitudinal variations in dyke geochemistry. In Fig. 8 there are suggestions of systematic trends within individual dykes of increasing P_2O_5 and decreasing MgO toward the north. Although each dyke trend is probably not statistically acceptable, it is significant that all five dykes that have been sampled longitudinally show similar trends.

Whether these trends are a subtle reflection of the southerly increase in hydrous alteration, which is known to increase MgO (Southwick and Day 1983), cannot yet be determined. Other possible explanations are that they arise from contamination by host rock or host-rock fluids that become less MgO rich toward the north, or from progressive fractionation of magma moving with a northward component of flow from a magmatic centre or "hot spot" beneath the Animikie Basin. Further geochemical work is required before these explanations or even the reality of the small longitudinal trends can be evaluated.

Conclusions

Using geochemical data as a correlation tool, it has been possible to demonstrate a characteristic remanence direction for individual dykes of the Kenora–Kabetogama dyke swarm over distances along strike exceeding 300 km. This consistency along strike, coupled with its variability between dykes, constitutes a definitive test for the primary nature of the remanence.

The use of geochemistry for long-distance correlation of dykes is possible because chilled margin samples provide diagnostic fingerprints if less mobile elements such as P, Ti, Zr, and Y are used. The results suggest that even where a dyke swarm is composed of a single magma type, there will be sufficient variation in abundances of these elements between dykes to permit longitudinal correlations.

Such long-distance correlations will help define variations in chemistry, paleomagnetism, and other properties along a dyke swarm that may reflect increasing proximity to continental margins active during and subsequent to dyke emplacement. In the KK swarm there are suggestions of increasing hydrous alteration, MgO content, and secondary magnetic overprinting toward such an active continental margin. The precise cause of these changes in terms of the rifting, sedimentation, and subsequent deformation of this margin during the period 2.2–1.6 Ga remains conjectural. However, there is no doubt that the overall uniformity in composition, the great lateral extent, and the short time span of dyke swarms, as exemplified by the KK dykes, suggest they are an important means to assess the geological history of cratonized Precambrian terrains adjacent to active continental margins.

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